

Venkat Ankit Gundala

(682) 374 7938 |gundalavenkatankit@gmail.com |LinkedIn

Professional Summary

Software Engineer with 4+ years of experience turning business and operational problems into reliable backend systems, automated workflows, and AI-enabled applications. Takes ownership from problem definition and system design through implementation, validation, deployment, and production support. Works across engineering, product, operations, and business teams to replace repetitive processes with scalable solutions, using AI-assisted and agentic methods while retaining human control over architecture, security, validation, and production decisions. Balances technical quality, delivery speed, system reliability, and business value.

Technical Skills

Languages: Java (8/11/17), Python, JavaScript, TypeScript, SQL, C++

Backend and APIs: Spring Boot, Spring MVC, Node.js, Express.js, Django REST Framework, Hibernate, Prisma, REST, GraphQL, gRPC

AI and Agentic Systems: Multi-Agent Orchestration, MCP Servers, Tool Calling, Structured Outputs, Workflow State, Human-in-the-Loop Approval, Prompt Engineering, RAG, Pinecone, FAISS

Databases and Messaging: MySQL, PostgreSQL, MongoDB, Redis, Oracle, Kafka, RabbitMQ

Cloud and DevOps: AWS (S3, EC2, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, GitHub Actions

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Testing and Tools: JUnit, Mockito, Pytest, Selenium, Postman, Git, Maven, Jira, Linux

Professional Experience

FoodSupply.ai, Software Engineer | USA

Jul 2024 – Present

- Architected a multi-agent inventory intelligence system with specialized agents for inventory investigation, demand forecasting, reorder analysis, policy retrieval, and operational reporting, separating responsibilities across each stage of the decision workflow.
- Built an MCP-based integration layer that exposed inventory, order, forecasting, and policy-retrieval services as typed agent tools, centralizing validation, role-based access, audit logging, retries, and human approval for reorder actions.
- Designed planning and task-decomposition flows that converted broad operational requests into sequenced steps for data retrieval, cross-store comparison, forecast evaluation, recommendation generation, and response validation.
- Implemented structured tool calling with typed input and output schemas, enabling agents to query inventory APIs, retrieve order history, invoke forecasting services, compare store performance, and prepare controlled reorder actions, accelerating order placement by 40%.
- Added session and workflow memory for conversational context, tool outputs, completed tasks, and pending actions, supporting operational queries grounded in a curated dataset of more than 15,000 inventory, order, shortage, and policy interactions.
- Introduced human-in-the-loop approval gates for reorder creation, inventory adjustments, and other high-impact actions, preventing autonomous changes to production data without operational review.
- Developed iterative planning, execution, observation, and correction loops with schema validation, retries, timeout handling, fallback paths, and audit logging, maintaining nightly reconciliation failures below 2%.
- Deployed backend services and AI workflows on AWS using EC2, Lambda, S3, and CloudWatch, supporting scheduled inventory processing, model-artifact storage, centralized logging, and production monitoring across forecasting and agent-driven workflows.
- Combined agentic workflows with retrieval-augmented generation and LSTM and SARIMA forecasting, achieving 85% forecast accuracy for high-volume SKUs, reducing expired inventory by 30%, and resolving more than 90% of pilot operational queries.

Accenture (Bank of America), Software Engineer | India

Nov 2021 – Jul 2022

- Developed a stateless Spring Boot rewards engine that classified card transactions and applied cashback rules based on customer tier, merchant category, spending behavior, and campaign period.
- Replaced hard-coded campaign logic with hot-reloadable YAML configurations, enabling product and marketing teams to introduce or modify cashback campaigns without application code changes or service downtime.
- Automated cashback assignment and customer-tier evaluation for more than 100,000 daily transactions, reducing processing latency from 2.1 seconds to under 700 milliseconds through caching and optimized rule execution.
- Replaced manual reward-discrepancy reviews with audit records, Spring AOP transaction logging, scheduled reconciliation, and Spring Batch reporting workflows for compliance and operations teams.

- Deployed Spring Boot services on AWS using EC2, SQS, CloudWatch, and CodeDeploy, enabling asynchronous reward processing, centralized operational monitoring, and consistent releases for high-volume cashback workflows.
- Added Redis-based idempotency, transactional rollback handling, asynchronous notification retries, and JUnit and Mockito coverage above 85% to prevent duplicate rewards and improve release reliability.

Radcube, Software Developer | India

Jun 2020 – Nov 2021

- Replaced merchant-specific SDK development with a configuration-driven payment SDK that accepted runtime UI, layout, and feature settings through APIs, reducing repeated coding and accelerating client onboarding.
- Created reusable payment modules and configurable checkout previews for clients including Zee5 and BookMyShow, allowing product and integration teams to validate payment experiences without rebuilding and redistributing SDK versions.
- Improved checkout responsiveness by pre-rendering static payment-page components during cart review and deferring sensitive payment-data retrieval until authorization.
- Built Django and GraphQL services for payment initiation, status tracking, transaction history, Stripe and PayPal integrations, and webhook synchronization, improving transaction success rates by 25%.
- Hosted Django payment services and background workers on AWS EC2, stored transaction and reporting artifacts in S3, and configured CloudWatch monitoring to improve deployment reliability, failure visibility, and recovery across payment-processing workflows.
- Optimized PostgreSQL queries, indexes, API payloads, and Dockerized worker services, reducing processing latency from approximately two seconds to under one second and increasing concurrent capacity by 50%.

Education

University of Texas at Arlington | MS in Computer Science

Aug 2022 – May 2024

Jawaharlal Nehru Technological University Hyderabad | BS in Computer Science

Jul 2017 – May 2021

Project Experience

AI Enhanced Code Review Platform | React.js, NestJS, PostgreSQL, OpenAI Codex

- Developed a full stack AI platform integrating OpenAI Codex to automate code reviews by suggesting best practice improvements for Python, Java, and JavaScript pull requests; implemented GitHub OAuth for seamless developer collaboration and feedback, significantly enhancing code quality and review efficiency.

End to End Text to SQL AI Platform | React.js, FastAPI, PostgreSQL, OpenAI API, AWS

- Built a full stack application enabling users to input natural language queries, converted them to optimized SQL statements using GPT powered LLMs, executed securely against a PostgreSQL database, and displayed results interactively, improving database accessibility and query efficiency.

Distributed Event Driven Microservices System | Java, Spring Boot, Kafka, Redis, PostgreSQL

- Implemented a scalable distributed microservices system using Kafka for event driven communication and gRPC for efficient inter service calls. Deployed microservices with Docker and managed orchestration via Kubernetes in an AWS cloud environment.